Seminar8

luni, 6 decembrie 2021 09:57

LR: $\cup, \cap, \circ, *, \setminus, C$

Proprietati de inchidere LIC : $\cup, \circ, *$

GIC, APS.

GIC: $L_1, L_2 \in LIC. \Rightarrow G_1 \to L_1$

U: $G_2 \to L_2.$

$G_1 = (V_1, \Sigma, R_1, S_1)$

$G_2 = (V_2, \Sigma, R_2, S_2)$

$a^n b^n$

$S \to aSb \mid e$

$L_1 = \{ a^n b^m / n > m \}$

$L_2 = \{ a^n b^m / n < m \}.$

G_1: $S \to aS_1 \mid aS$

$S_1 \to aS_1b \mid e$

G_2: $S \to S_1b \mid Sb$ (bS_1 | bS)

$S_1 \to aS_1b \mid e$

$S \to bS_1 \Rightarrow baS_1b$

$L_3 = L_1 \cup L_2 = \{ a^n b^m / n > m \text{ sau } n < m \}$

$= \{ a^n b^m / n \neq m \}.$

$S \to aSb \mid S_1 \mid S_2$ $\qquad$ $aaab$

$S_1 \to aS_1 \mid a$

$S_2 \to bS_2 \mid b$

$S \to S_{11} \mid S_{21}$

$$\begin{cases} S_{11} \to a S_{12} \mid a S_{11} \\ S_{12} \to a S_{12} b \mid e \end{cases}$$

$$\begin{cases} S_{21} \to S_{22} b \mid S_{21} b \\ S_{22} \to a S_{22} b \mid e \end{cases}$$

$(V_1 \setminus \Sigma_1) \cap (V_2 \setminus \Sigma_2) = \emptyset$

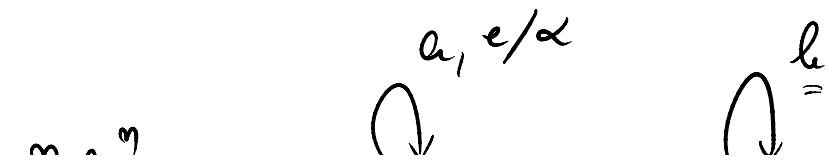
$G_3 = (V_1 \cup V_2 \cup \{S\}, \Sigma, R_1 \cup R_2 \cup \{(S, S_1), (S, S_2)\}, S)$

S símb. de start al G_1 ; S símb. de start al G_2.

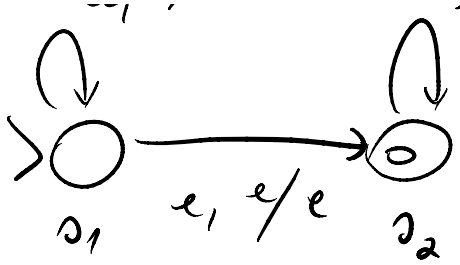
APD.

$L_1 = \{ a^n b^m / n > m \}$

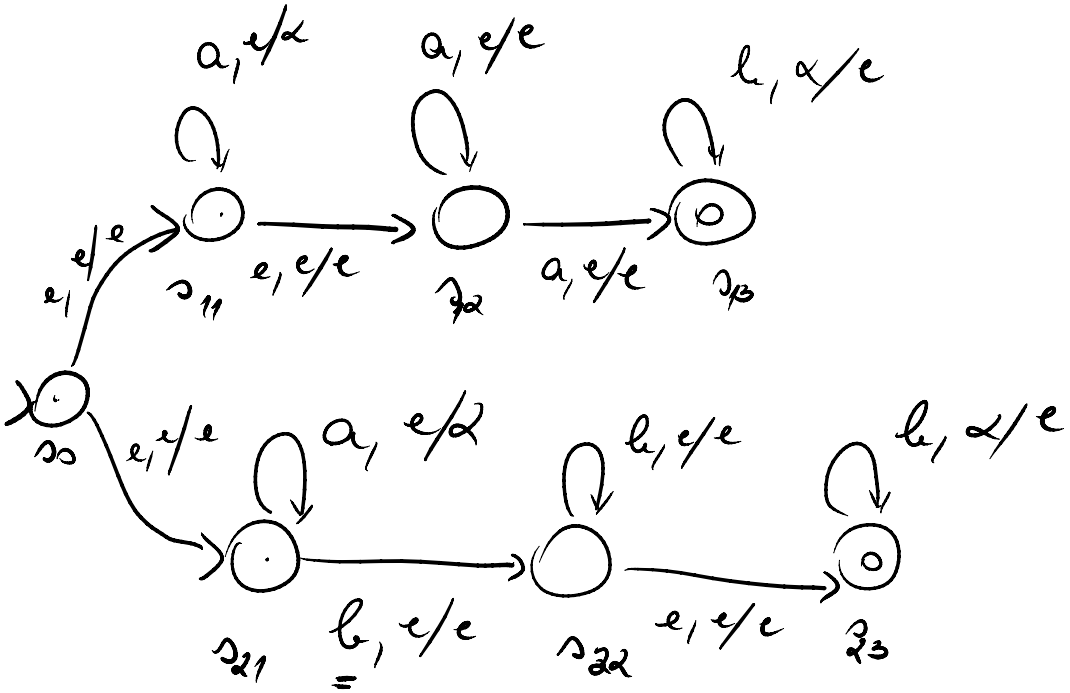
$L_2 = \{ a^n b^m / n < m \}$.



$a^n b^n$



1. am "e" la intrare
2. sunt în stare finală
3. stiva e goală



$K_1 \cap K_2 = \emptyset$

$M_1 = (K_1, \Sigma, \Gamma_1, \Delta_1, s_1, F_1)$

$M_2 = (K_2, \Sigma, \Gamma_2, \Delta_2, s_2, F_1)$

$M_3 = (K_1 \cup K_2 \cup \{s_0\}, \Sigma, \Gamma_1 \cup \Gamma_2, \Delta_1 \cup \Delta_2$

$\cup \{((s_0, \varepsilon, \varepsilon), (s_1, \varepsilon)), ((s_0, \varepsilon, \varepsilon), (s_2, \varepsilon))\},$

$s_0, F_1 \cup F_2)$

$\Rightarrow G_1 \to L_1 \quad , G_2 \to L_1 \cup L_2$

$L_1, L_2 \in LIC$ → $G_1 \to L_1$, $G_2 \to L_2$ → G_3 → $L_1 \cup L_2$

→ $M_1 \to L_1$, $M_2 \to L_2$ → M_3 → $L_1 \cup L_2$.

$\forall\ L_1, L_2 \in LIC \Rightarrow L_1 \cup L_2 \in LIC$

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prop. de închidere

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o :

$S_{11} \to a\,S_{12} \mid a\,S_{11}$ } → G_1

$S_{12} \to a\,S_{12}\,b \mid e$

$S_{21} \to S_{22}\,b \mid S_{21}\,b$ } → G_2.

$S_{22} \to a\,S_{22}\,b \mid e$

$S \to S_{11}\,S_{21}$

$L_3 = L_1 \circ L_2$.

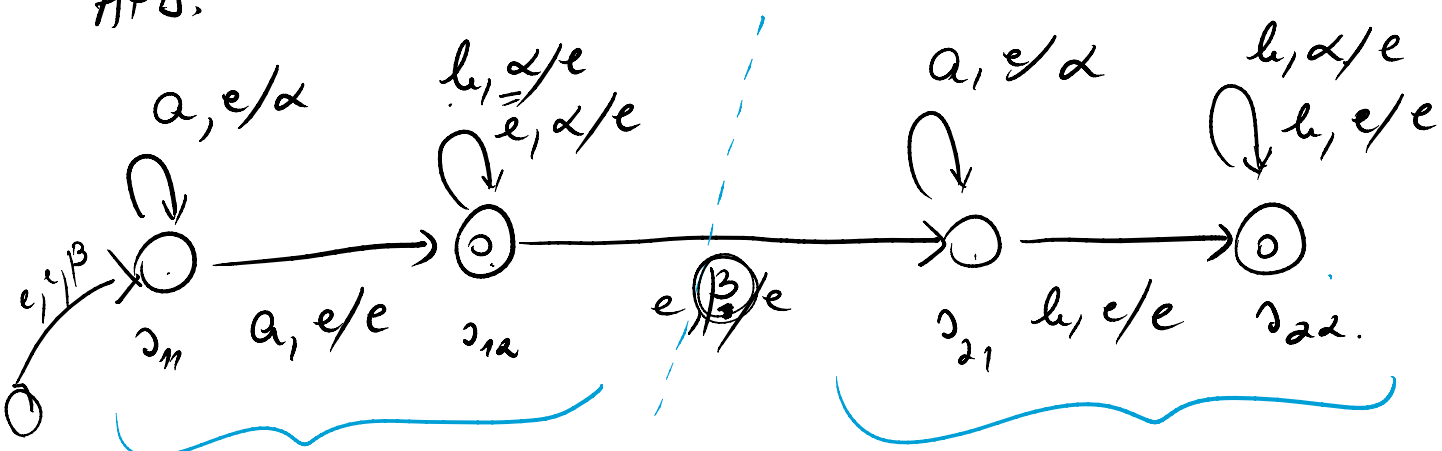
$L_1 = \{ a^n b^m / n > m \}$

$L_2 = \{ a^n b^m / n < m \}$

$$L_3 = L_1 \circ L_2 = \{ a^{n_1} b^{m_1} a^{n_2} b^{m_2} / n_1 > m_1 \; n_2 < m_2 \}$$

$$G_3 = (V_1 \cup V_2 \cup \{S\}, \Sigma, R_1 \cup R_2 \cup \{(S, S_1 S_2)\}, S)$$

APD.



M_1

stiva e goală

M_2.

aabbabb

$|x|_1 = |x|_2$

$$\Gamma_1 \cap \Gamma_2 = \emptyset$$

$M_3 = (K_1 \cup K_2, \Sigma, \Gamma_1 \cup \Gamma_2, \Delta_1 \cup \Delta_2 \cup$

$\{((q, e, e), (s_2, e)) / q \in F_1, q_1, s_1,$

$F_2)$.

$L_1 \circ L_2 \begin{cases} \xrightarrow{GIC} \\ \xrightarrow[APD]{} \end{cases} \Rightarrow L_1 \circ L_2 \in LIC.$

$\Rightarrow \forall L_1, L_2 \in LIC \Rightarrow L_1 \circ L_2 \in LIC.$

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*: $L_1 \quad S_1 \to aS_1 b / aS_1 / a.$

$a^n b^m / n > m \quad \underline{S} \to \underline{SS_1} / \underline{e}.$

$L_1^* = \{ a^{n_1} b^{m_1} a^{n_2} b^{m_2} \ldots a^{n_k} b^{m_k} /$

$n_i > m_i \}$

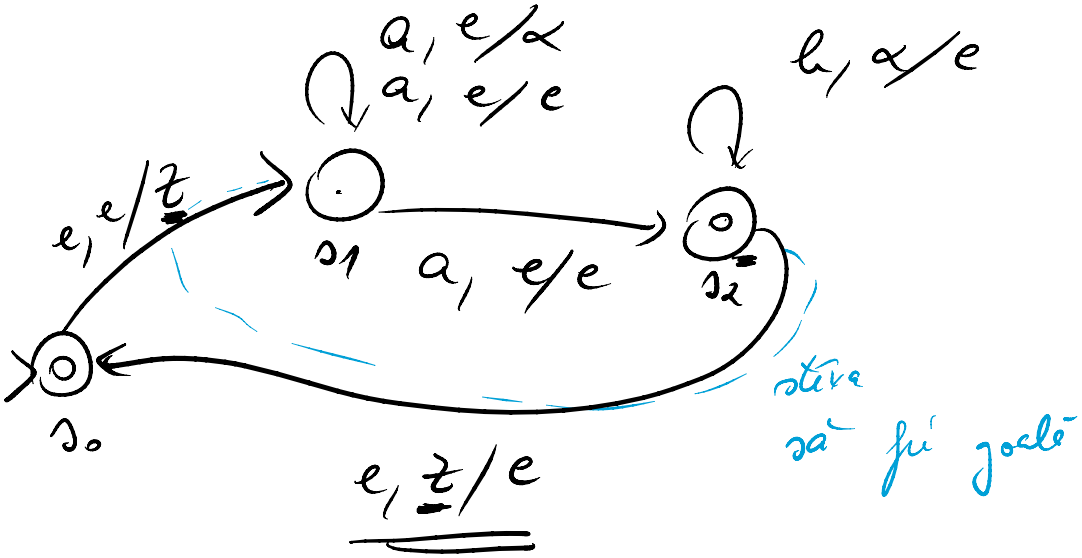
$e \in L_1^*$

$S \Rightarrow S S_1 \Rightarrow S S_1 S_1 \Rightarrow S S_1 S_1 S_1 \ldots$

$G_3 = (V_1 \cup \{S\}, \Sigma, R_1 \cup \{(S, SS_1), (S, \varepsilon)\}, S)$

APD.

$a^n b^m, \quad n \geq m$



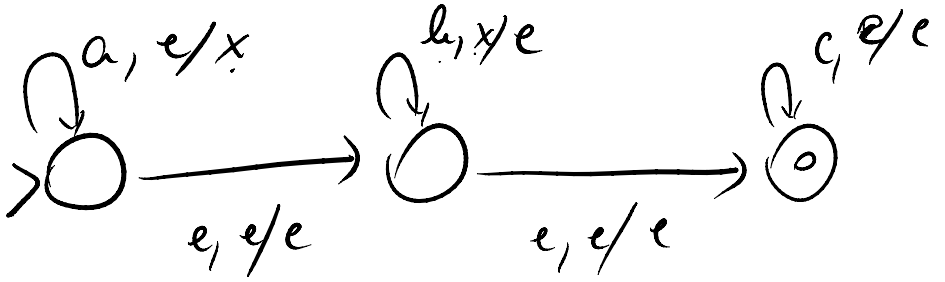
$M_3 = (K_1 \cup \{s_0\}, \Sigma, \Gamma_1 \cup \{z\}, \Delta, \cup$

$\{((s_0, e, e), (s_1, z)), ((f, e, z), (s_0, e)) / f \in F_1\}, s_0, \{s_0\})$

L_1^* GIC

APD $\Rightarrow \forall L_1 \in LIC \Rightarrow L_1^* \in LIC.$

$\cap$: $L_1 = \{ a^m b^n c^k / m = n \} \in LIC.$



$L_2 = \{ a^m b^n c^k / n = k \} \in LIC$

$L_1 \cap L_2 = \{ a^m b^n c^k / \begin{matrix} m = n \\ n = k \end{matrix} \}.$

$a^n b^n c^n \notin LIC$

$L_1 \setminus L_2 = \{ a^m b^n c^k / m = n, n \neq k \}$

$\notin LIC.$

$L = \{ !w\tilde{w} / w \in \Sigma^* \}. \in LIC.$

$w = w_1 w_2$, $\#_{w_1} = \#_{w_2}$.

$w_1 \neq w_2$.

$$C_L = \{ ww \mid w \in \Sigma^* \} \notin LIC.$$

$$\underbrace{aabc}_{w_1} \mid \underbrace{aabc}_{w_2}$$

$\forall L_1 \in LIC$ y $L_2 \in LR$.

$\Rightarrow L_1 \cap L_2 = L_3 \in LIC$.

$\forall L_1 \in LIC$ y $L_2 \in LR$

$L_1 \setminus L_2 = L_3 \in LIC$.